



**D.P.U. 24-196: Comments of the Vehicle-Grid Integration Council (VGIC)
on February 5, 2025 EV Infrastructure Program Hearing**

I. Introduction

The Vehicle-Grid Integration Council (VGIC)¹ is a 501(c)(6) membership-based trade association committed to advancing the role of electric vehicles (EVs) and vehicle-grid integration (VGI) through policy development, education, outreach, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring the value from flexible EV charging and discharging is recognized and compensated in support of a more reliable, affordable, and efficient electric grid. VGIC appreciates the opportunity to provide comments to the Department of Public Utilities (DPU) on the February 5 2025 EV Infrastructure Program Hearing.

II. Vehicle-to-Grid Bidirectional Charging Systems Provide Critical Benefits to Massachusetts EDC Ratepayers.

a. Acceleration of Transportation Electrification

Vehicle-to-grid (“V2G”) bidirectional charging solutions can significantly reduce the total cost of ownership for electric vehicles (EV), including both passenger vehicles and EV fleets, by enabling participation in grid services programs. A pertinent example is the Connected Solutions program, which offers incentives for demand response participation. Such performance-based incentives can be key drivers for fleets considering the transition to EVs.

b. Support for the Evolving Power Sector

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., BorgWarner, Fermata Energy, Ford Motor Company, General Motors, Nissan North America, Bidirectional Energy, dcbel, Eaton, Emporia, EnergyHub, ev.energy, Gravity, GridBeyond, GridWiz, Hyundai Cradle, Kaluza, Landis and Gyr, Mercedes-Benz, Orange Charger, Peak Power, EDF Powerflex, Stellantis N.V., Sumitomo Electric, Sunrun, Tesla, The Mobility House, Toyota Motors North America, WeaveGrid, Clean Power Alliance, Holy Cross Energy, Los Angeles Department of Water and Power, and Sacramento Municipal Utility District. The views expressed in these comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>).

V2G facilitates the integration of renewable energy sources by providing necessary energy storage and grid services. By enabling bidirectional energy flow at the grid edge, V2G can assist in balancing supply and demand, thereby enhancing grid stability and accommodating higher penetrations of renewable energy.

c. Downward Pressure on Electric Utility Rates

Efficient utilization of existing infrastructure through V2G can alleviate distribution congestion and defer the need for costly infrastructure upgrades. Implementing V2G as a non-wires alternative (NWA) in areas with distribution constraints can lead to more efficient grid operations and exert downward pressure on utility rates. V2G resources can also offset the use of costly peaking capacity that may otherwise be procured in the ISO-NE market.

d. Enhancement of Resilience

V2G technology can provide critical backup power during emergencies. For instance, electric fleet vehicles equipped with V2G capabilities can supply electricity to evacuation centers or other essential facilities during power outages, thereby bolstering community resilience. On the residential side, passenger EVs can provide home backup power to customers during planned or unplanned outages.

e. Fostering Economic Activity

Many of VGIC's members and supporters have a presence in Massachusetts, including offices, manufacturing facilities, dealerships, and/or technical personnel. The advancement of V2G technology is poised to stimulate economic growth within the Commonwealth by creating jobs and fostering innovation in both the clean energy and transportation sectors.

III. Barriers to Widespread V2G Implementation Persist Despite Massachusetts' Recent Progress.

VGIC acknowledges the progress made in Massachusetts toward facilitating widespread V2G. Notably, the Connected Solutions program provides performance-based compensation for customers utilizing bidirectional charging systems, and interconnection with utility distribution grids as a form of energy storage is permitted by EDCs, including an exemption from UL 1741 SB for certain bidirectional charging equipment. Additionally, V2G equipment and software are currently available from major school bus manufacturers and charging equipment providers,

including Fermata Energy, BorgWarner, The Mobility House, Tellus Power Green, InCharge, Heliox, and Synop.

Despite these advancements, significant barriers persist that impede the widespread adoption of V2G technology. Foremost among these is the substantial upfront cost associated with deploying V2G bidirectional charging solutions compared to unidirectional charging systems. Presently, existing energy storage technology deployment incentives do not recognize bidirectional charging systems as eligible, and the current slate of utility EV programs lack incremental incentives specific to this technology. This absence of financial support significantly hinders market transition toward V2G bidirectional charging solutions, which may need this support only while in the relatively early stages of market development.

To overcome these challenges, VGIC strongly recommends the implementation of targeted upfront incentives to offset the initial costs of bidirectional charging infrastructure. Such incentives are essential to accelerate market adoption and fully realize the potential benefits of V2G technology within Massachusetts. VGIC respectfully urges the DPU to allow these incentives to be claimed by projects that may claim other federal or state incentives for this technology, so as to not preclude projects from successful implementation or overly restrict access to funding intended to support market transition for the relatively nascent V2G space.

IV. VGIC Supports Eversource's Proposed Incremental V2G Charger Incentive for Electric School Bus Sites and Recommends Additional Actions for the EDCs to Support V2G.

VGIC commends Eversource for its recognition of the value of V2G technology in supporting electric school buses and strongly supports this element of its proposal. Establishing incremental incentives for V2G bidirectional charging infrastructure at electric school bus charging sites aligns with emerging national best practices, as seen in California and under early consideration in other states like New York, Maryland, and Illinois.

To ensure that Massachusetts remains at the forefront of vehicle-grid integration, VGIC urges the DPU to direct National Grid and Unitil to follow Eversource's lead by offering incrementally higher incentives for bidirectional charging equipment relative to unidirectional charging systems. A consistent and equitable approach across all three investor-owned electric distribution companies (EDCs) will be critical in fostering a robust and competitive market for V2G solutions.

Further, while Eversource's proposal focuses on the electric school bus sector, VGIC strongly recommends that all three EDCs develop bidirectional charging incentives beyond this

single use case. Residential managed charging programs and medium- and heavy-duty fleet applications other than electric school bus charging also stand to benefit from bidirectional capabilities, delivering increased grid benefits, improved reliability, and cost savings for both participants and non-participating ratepayers. Expanding the scope of V2G incentives will ensure that Massachusetts fully leverages the potential of this technology across a broad set of use cases.

VGIC also urges the DPU to direct all EDCs to implement additional performance-based programs and/or optional export compensation rates that are available to a broader set of customers, including residential customers with V2G equipment. While programs such as Connected Solutions have provided an important pathway for fleet and commercial V2G participation, residential customers remain largely excluded from V2G compensation opportunities. Ensuring that households with bidirectional EVs can access fair compensation for grid exports will further support market growth and enhance grid flexibility.

Finally, VGIC reiterates its recommendation that all incremental V2G incentives remain eligible to be claimed by sites that may also receive other state or federal-funded incentives. Layering incentives will be essential to overcoming the high upfront costs of bidirectional charging infrastructure and accelerating adoption, especially as the current V2G market is relatively nascent. By adopting these recommendations, the DPU can position Massachusetts as a leader in V2G deployment, delivering benefits to ratepayers, advancing grid resilience, and supporting the state's clean energy and transportation electrification goals.

V. Conclusion

VGIC appreciates the opportunity to provide these comments and looks forward to working with Eversource, National Grid, Unitil, the Department, and other stakeholders to ensure the success of Massachusetts's transportation electrification efforts.

Respectfully submitted,

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